Advanced statistics

21-01-26

categorical data

A - 45 B - 32

probability function $= f(x) = p^{x}(1-p)^{1-x}$

where $p = \frac{45}{77}$

$f(1) = \frac{45}{77}$ $f(0) = 1 - \frac{45}{77} = \frac{32}{77}$

mathematical representation of data distribution

$$\boxed{f(x) = p^{x}(1-p)^{1-x}}$$

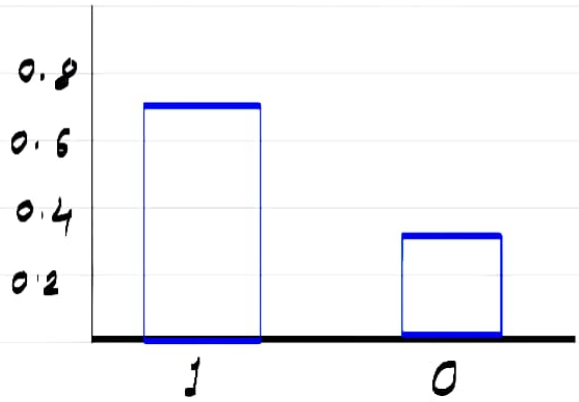
this is known as Bernouli distribution

Bernouli Distribution

- used for representing data distribution of categorical data having 2 categories
- Its also associated to a random experiment where exactly two outcomes are possible.
- Its also used in machine learning in binary classification models.

$$f(x) = p^x (1-p)^{1-x}$$

$$p = 0.7$$



random experiments related to Bernouli dist

1) Tossing a coin

sample space = {H, T}

$$f(x) = p^x (1-p)^{1-x}$$

$$p = 1/2 = 0.5$$

$p(H) = f(1) = 0.5$ $\quad$ $p(T) = f(0) = 0.5$

2) A bucket contains 40 black and 25 white balls
A ball is randomly taken out from bucket.

sample space = {black, white}

$$f(x) = p^x (1-p)^{1-x}$$

$$p = \frac{40}{65}$$

$p(\text{black}) = f(1) = \frac{40}{65}$ $\quad$ $p(\text{white}) = f(0) = \frac{25}{65}$

- A modified version of Bernouli dist gives us the Binomial distribution

Binomial distribution

- Its an extension of bernouli dist

Random experiment - tossing a coin

lets say H is success = 1 $p = 0.5$

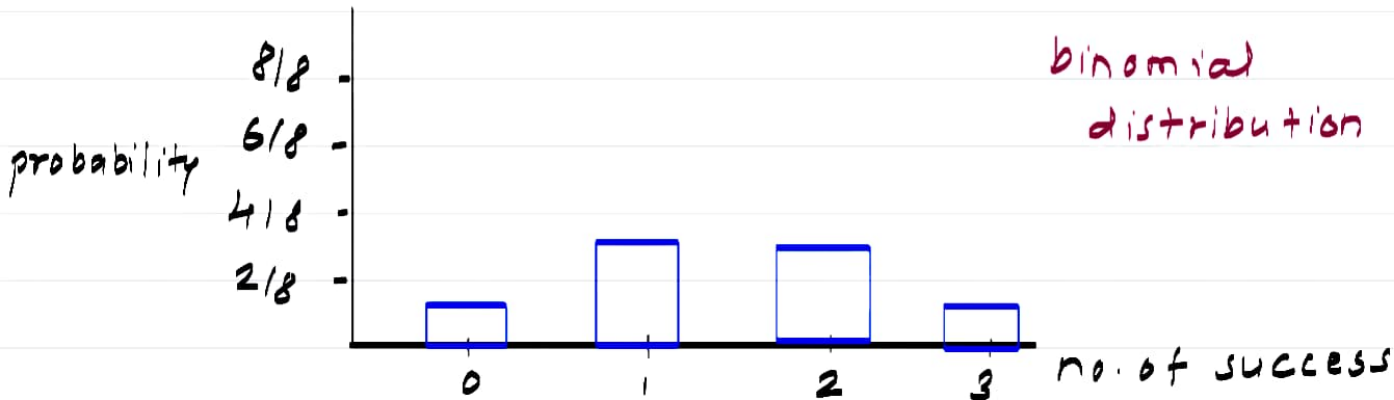
3 times → sample space = { HHH, HTH, THH, HHT, TTT, THT, HTT, TTH }

2 times → sample space = { HH, HT, TH, TT }

1 time → sample space = { H, T }

tossing a coin 3 times

probability of exactly 0 success = 1/8
probability of exactly 1 success = 3/8
——"—— 2 successes = 3/8
——"—— 3 successes = 1/8
4 ——"—— = 0
5 ——"—— = 0



- bernouli dist is related to bernouli trial
- bernouli trial is random experiment which has only 2 possible outcomes
- bernouli dist gives probability of success

Tossing a coin once

$SS = \{H, T\}$

$P(\text{success}) = 0.5$

- binomial trial is simply the bernouli trial done multiple times.
- binomial dist gives probability of k successes in n trials

Tossing a coin 3 times

$SS = \{HHH, HTH, HHT, THH, TTT, THT, TTH, HTT\}$

prob of 0 success in 3 trials $= 1/8 = P(k=0, n=3)$

" 1 " " $= 3/8 = P(k=1, n=3)$

" 2 " " $= 3/8 = P(k=2, n=3)$

" 3 " " $= 1/8 = P(k=3, n=3)$

for binomial distribution below is the probability function

$$\boxed{P(k, n) = {}^{n}C_{k}\, p^{k} (1-p)^{n-k}}$$

where p is Probability of success in single trial

Random experiment with only 2 possible outcomes

↓

bernouli trial

single trial → bernouli distribution

multiple trials → binomial distribution

binomial dist

$$P(k,n) = {}^{n}C_{k}\, p^{k}(1-p)^{n-k}$$

where

$p \rightarrow$ prob. of success in single trial

n - no. of trials

k - no. of successes

$n! = 1 \times 2 \times 3 \times 4 \cdots (n-1) \times n$

$1! = 1$

$2! = 1 \times 2 = 2$

$0! = 1$

$3! = 1 \times 2 \times 3 = 6$

$4! = 24$

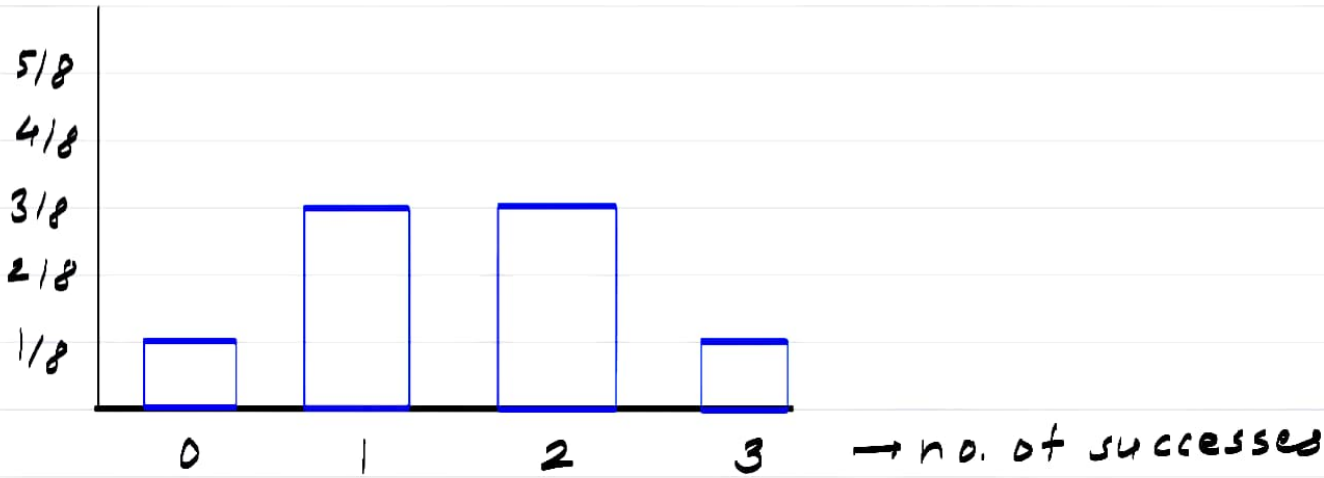
$5! = 120$

binomial coefficient ← $\boxed{{}^{n}C_{k} = \dfrac{n!}{k!\,(n-k)!}}$

$${}^{4}C_{2} = \frac{4!}{2!\,(4-2)!} = \frac{24}{2 \times 2} = 6$$

$${}^{3}C_{1} = \frac{3!}{1!\,(3-1)!} = \frac{6}{1 \times 2} = 3$$

tossing a coin 3 times



$$\left[p(n,k) = {}^{n}C_{k}\, p^{k} (1-p)^{n-k} \quad \text{where } p = 0.5 = 1/2 \right]$$

$$p(n=3, k=0) = {}^{3}C_{0} \left(\frac{1}{2}\right)^{0} \left(1 - \frac{1}{2}\right)^{3-0}$$

$$= \frac{3!}{0!\,3!} \times 1 \times \left(\frac{1}{2}\right)^{3}$$

$$p(n=3, k=0) = \frac{1}{8}$$

$$p(n=3, k=2) = {}^{3}C_{2} \left(\frac{1}{2}\right)^{2} (1 - 1/2)^{3-2}$$

$$= \frac{3!}{2!\,1!} \times \frac{1}{4} \times \frac{1}{2}$$

$$= \frac{6}{2} \times \frac{1}{8} = \frac{3}{8}$$

$$p(n=3, k=2) = 3/8$$

A bucket has 40 white and 30 black balls
lets say white — success

Random experiment - taking a ball out of this bucket randomly

$p(\text{success}) = p(\text{white}) = \frac{40}{70} = \frac{4}{7}$

Random experiment — taking 'n' balls out of this bucket

lets say $n = 10$

prob. of getting exactly 4 white balls $= {}^{n}C_{k}\, p^{k}(1-p)^{n-k}$

$n = 10$
$k = 4$
$p = 4/7$

$p(n=10, k=4) = {}^{10}C_{4} \left(\frac{4}{7}\right)^{4} \left(1-\frac{4}{7}\right)^{10-4}$

prob. of getting exactly 6 white balls $= {}^{10}C_{6} \left(\frac{4}{7}\right)^{6} \left(1-\frac{4}{7}\right)^{10-6}$

prob. of getting exactly 2 black balls $= {}^{10}C_{2} \left(\frac{3}{7}\right)^{2} \left(1-\frac{3}{7}\right)^{10-2}$

↕ same values

↓

8 white balls $= {}^{10}C_{8} \left(\frac{4}{7}\right)^{8} \left(1-\frac{4}{7}\right)^{10-8}$